

 ALBUKHARY INTERNATIONAL UNIVERSITY	SCHOOL OF COMPUTING AND INFORMATICS CCE2333- Software Testing
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Module 2: Boundary Value Analysis & Equivalence Partitioning in Software Testing (Report)

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Group:	C

Marks:

Part A	Part B	Part C	Pard D

Program 1 — University Grade Classification

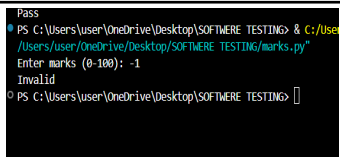
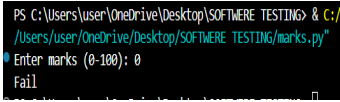
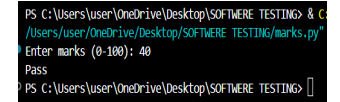
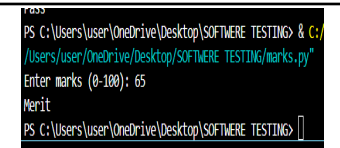
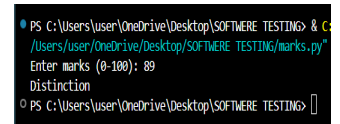
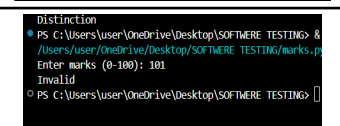
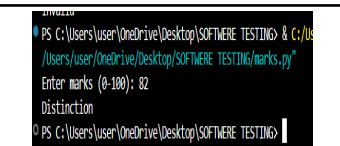
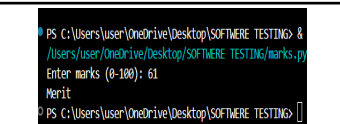
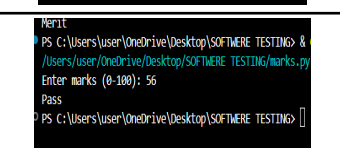
Part A1: Equivalence Classes (EP) Table

Class	Description	Range
EC1	the output is invalid as it is below the range	marks<0
EC2	Fail because the range between 0-39	0<marks <39
EC3	Pass as the range in 40 to 59	40<marks <59
EC4	Merit as the result in between the 60-70	60<marks <79
EC5	Distinction as the result between 80-100	80<marks<100

Part B1: Boundary Value Analysis (BVA) Table

Boundary	Test Value
lower fail boundary	-1,1,0
pass	39,40,41
merit	59,60,61
Distinction	79,80,81
invalid boundary	99, 100,101

Part C1: Test Cases & Results

	Test ID	Input	Expected Output	Actual Output	Pass/Fail
	TID 01	-1	INVALID		PASS
	TID 02	0	Fail		Pass
	TID 03	40	pass		Pass
	TID 04	65	merit		pass
	TID 05	89	Distinction		pass
	TID 06	101	Invalid		pass
	TID 07	82	Distinction		pass
	TID 08	61	merit		pass
	TID 09	56	Pass		pass

	TID 10	19	Fail	<pre> Merit PS C:\Users\user\OneDrive\Desktop\SOFTWARE TESTING> & .\marks.py Enter marks (0-100): 19 Fail PS C:\Users\user\OneDrive\Desktop\SOFTWARE TESTING> </pre>	pass

Program B - Discount Calculation

Part A1: Equivalence Classes (EP) Table

Class	Description	Range
EC1	this is invalid	amount<0
EC2	0% discount in this range	0-99
EC3	5 % discount in this range	100-499
EC4	10% discount in this range	500-999
EC4	20% discount in this range	amount >1000

Part B1: Boundary Value Analysis (BVA) Table

Boundary	Test Value
invalid boundary	-1,0,1
0%	99,100,101
5 % discount	499, 500,501
10% discount	999,1000,1001
20% discount	1000,10001,1002

Part C1: Test Cases & Results

Test ID	Input	Expected Output	Actual Output	Pass/Fail
TD01	-1	invalid	<pre> C:\Users\user\OneDrive\Desktop\SOFTWARE TESTING> .\main.py Enter total purchase amount (RM): -1 Invalid PS C:\Users\user\OneDrive\Desktop\SOFTWARE TESTING> </pre>	pass

TD02	0	0% Discount	<pre>e "c:/Users/user/OneDrive/Desktop/SOFTWARE TESTING/ Enter total purchase amount (RM): 0 0% Discount PS C:\Users\user\OneDrive\Desktop\SOFTWARE TESTING></pre>	pass
TD03	99	0% Discount	<pre>e "c:/Users/user/OneDrive/Desktop/SOFTWARE TESTING/hello.py Enter total purchase amount (RM): 99 0% Discount PS C:\Users\user\OneDrive\Desktop\SOFTWARE TESTING></pre>	pass
TD04	100	5% Discount	<pre>e "c:/Users/user/OneDrive/Desktop/SOFTWARE TESTING/hello.py Enter total purchase amount (RM): 100 5% Discount PS C:\Users\user\OneDrive\Desktop\SOFTWARE TESTING></pre>	pass
TD05	499	5% Discount	<pre>e "c:/Users/user/OneDrive/Desktop/SOFTWARE TESTING/hello.py Enter total purchase amount (RM): 499 5% Discount PS C:\Users\user\OneDrive\Desktop\SOFTWARE TESTING></pre>	pass
TD06	500	10% Discount	<pre>e "c:/Users/user/OneDrive/Desktop/SOFTWARE TESTING/hello.py Enter total purchase amount (RM): 500 10% Discount PS C:\Users\user\OneDrive\Desktop\SOFTWARE TESTING></pre>	pass
TD07	999	10% Discount	<pre>Enter total purchase amount (RM): 999 10% Discount PS C:\Users\user\OneDrive\Desktop\SOFTWARE TESTING></pre>	pass
TD08	1000	20% Discount	<pre>e "c:/Users/user/OneDrive/Desktop/SOFTWARE TESTING/hello.py Enter total purchase amount (RM): 1000 PS C:\Users\user\OneDrive\Desktop\SOFTWARE TESTING></pre>	fail
TD09	1200	20% Discount	<pre>e "c:/Users/user/OneDrive/Desktop/SOFTWARE TESTING/hello.py Enter total purchase amount (RM): 1200 20% Discount PS C:\Users\user\OneDrive\Desktop\SOFTWARE TESTING></pre>	pass
TD10	550	10% Discount	<pre>Enter total purchase amount (RM): 550 10% Discount PS C:\Users\user\OneDrive\Desktop\SOFTWARE TESTING></pre>	pass

Part D: Analysis

Program A

Programme is ok , the output is as expected , there is no error in the code , programme is working well and the logic flaw is also defined

Program B

Boundary value(s) that exposed the fault: programme B has a fault as the output is not expected and there is logic error in the code

Suggestion to fix the bug

For Solving the bug its need to change the code from *elif amount > 1000: this to elif amount >= 1000:*

Part E: Reflection Questions

Q1: How did applying BVA and EP help you identify edge cases that might have been missed with random testing?

BVA and EP ensure that edge cases and representative values from each partition are tested, reducing the chance of missing critical input ranges compared to random testing

Q2: If the fault in Program B existed in a real online store, what potential impact could it have on business operations and customer trust?

there will be a financial loss and loss of customer trust as the customer will not get the discount as expected

Part F: Reflection Questions

Summarize your understanding of BVA & EP:

BVA focuses on testing at the edges of input ranges where testers can find bug usually while EP divides inputs into valid/invalid classes to minimize redundant testing and ensure coverage.

-End of Module 2-